RFI Report for October 2024

Executive summary

From the samples we collected during the month of October 2024, the antennas at/around the core of MeerKAT are more affected by noise with comb-like patterns overall, as evidenced by the auto and short baseline cross correlation plots (see subsection a, subsection b, subsection c, subsection d). Another important finding is the RFI emitted by Telkom downlink, as can be seen on the plots related to SB_ID:20220801-0021 dataset observed on the 18th of October 2024 (see subsection c). The sub-band affected by Telkom downlink is not supposed to be used on site according to the RFI team.

In summary, the observed hourly RFI for October is moderate, with an overall average of 12%. However, significant spikes in RFI levels exceeding 60% were detected at 13:00 UTC and 20:00 UTC. These elevated RFI levels occur at known UHF frequencies, particularly between 544–592 MHz, associated with DTV1. Notably, this strong RFI was not present in all observations.

1. Introduction

The purpose of this document is to report the current status of RFI on-site. Given the construction work that has been going on over the last couple of months, it is therefore imperative to monitor the construction related activities on site that could potentially generate RFI signals which, in turn, contaminate the science observations. Preferably these reports are to be released on a weekly basis (or every two weeks).

2. Method

The flags data, specifically *ingest_flags*, are considered in these analyses, as they can be used as proxies for RFI frequency occupancy, i.e. RFI emission as a function of frequency channels. The strategy is to select two to three observations per week and report the status of the RFI contamination on site based on the flags of each dataset (of each observation). For each observation

- Autocorrelation flags of a few *scans* corresponding to the *gain calibrator* are selected. This is due to the fact that, at the time of writing, there are some issues with accessing a big chunk of an entire dataset.
- Autocorrelation flags, which are three dimensional arrays, i.e. *time*: *frequency channels*: *correlation products*, are averaged over both time and correlation products.
- Autocorrelation flags become a one dimensional array which is plotted in order to check the RFI occupancy in the frequency channels.

The cross correlation flags are also investigated in order to see how much the RFI emission affects the correlation products which are the main interest in an interferometric observation. The procedure for the cross correlation flags is the same as that of autocorrelation ones. To

further go into more detail, the flags corresponding to short and long baselines are split into two parts, and each part is plotted separately. Based on some datasets that were analyzed previously, the U-band observations are the ones that are most affected by the current RFI emissions on site. Hence, the majority of the datasets that will be inspected is in U-band, nevertheless observations in both L and S bands will be looked at too.
In what follows, it is worth noting that the vertical gray shaded areas in all plots denote the known RFI emissions in U-band.

3. Results

a. 2024, 04 October

The plots in Figures 1, 2 and 3 are obtained from a dataset collected on the 4th of October 2024. The autocorrelation is affected by relatively weak interferences with a fraction of time flagged value less than 0.1, especially on the lower part of the band, as evidenced in Figure 1. The short baseline cross correlation exhibits similar behavior with a slightly less number of contaminated channels (see Figure 2). Interestingly, the long baseline cross correlation appears to be relatively clean as indicated in Figure 3. This suggests that the short baseline antennas are the most affected, potentially owing to some RFI emitting activities at the core of the array.

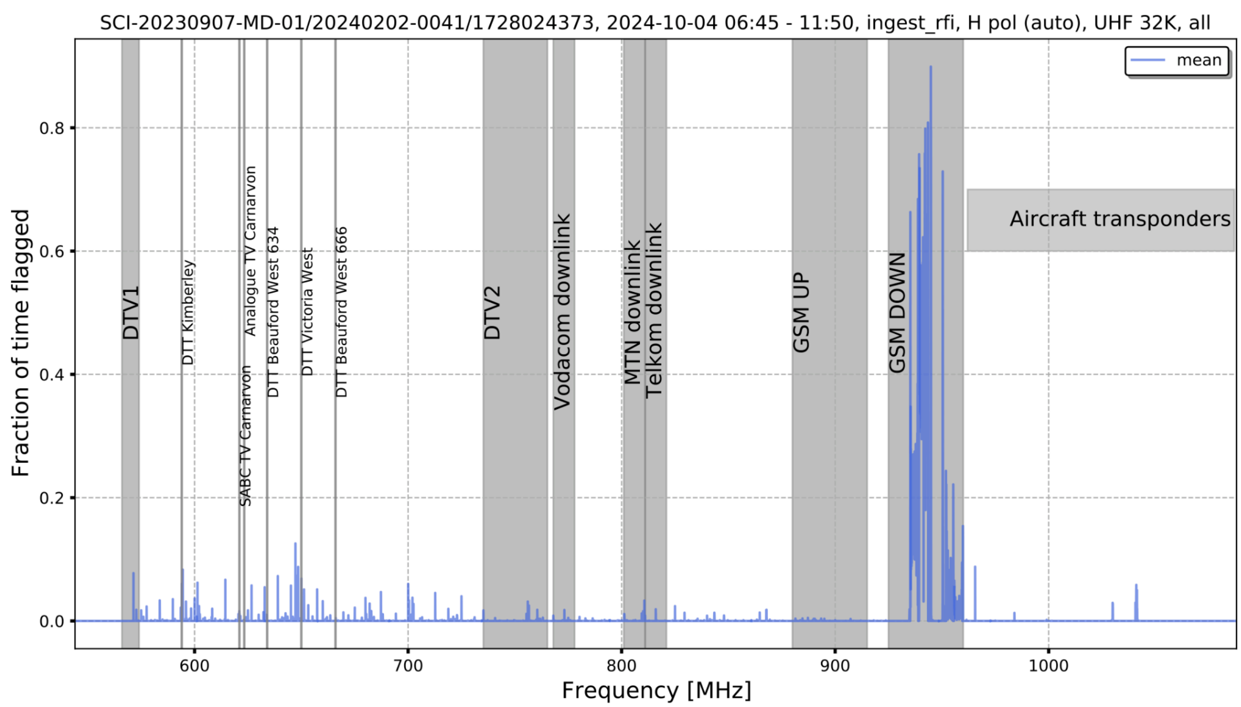

Figure 1: Mean of the autocorrelation flags of SB_ID:20240202-0041 dataset.

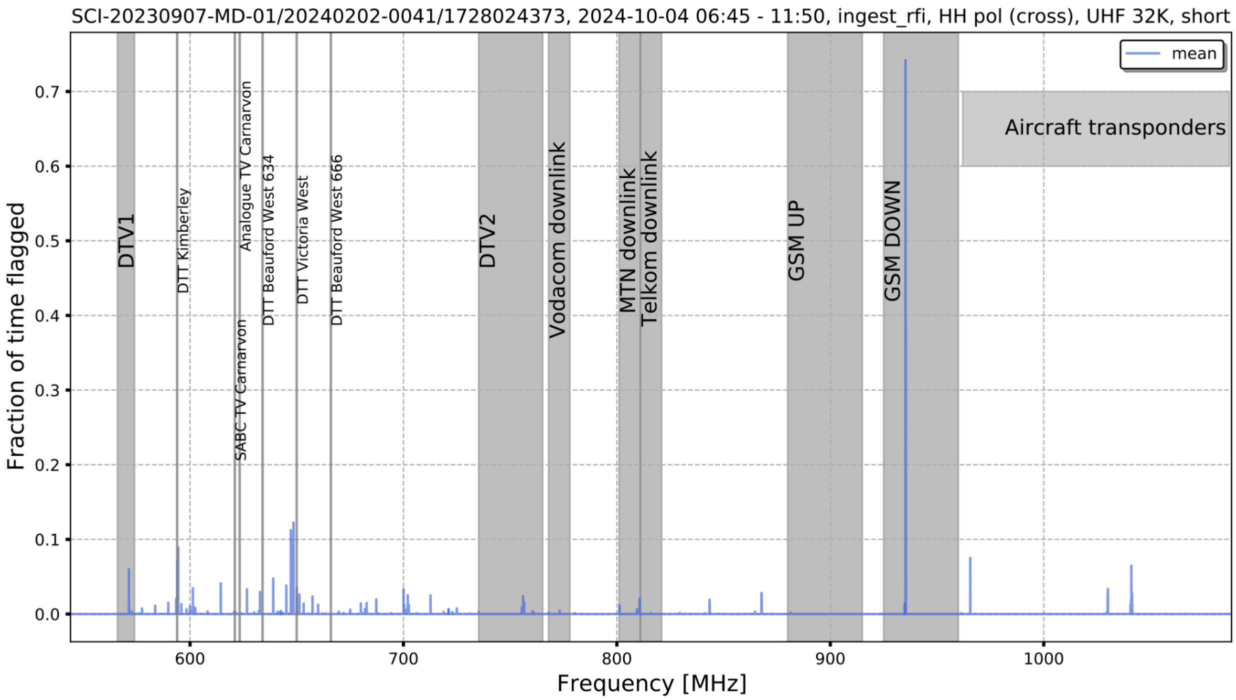

Figure 2: Mean of the cross correlation flags of the short baselines of SB_ID:20240202-0041 dataset.

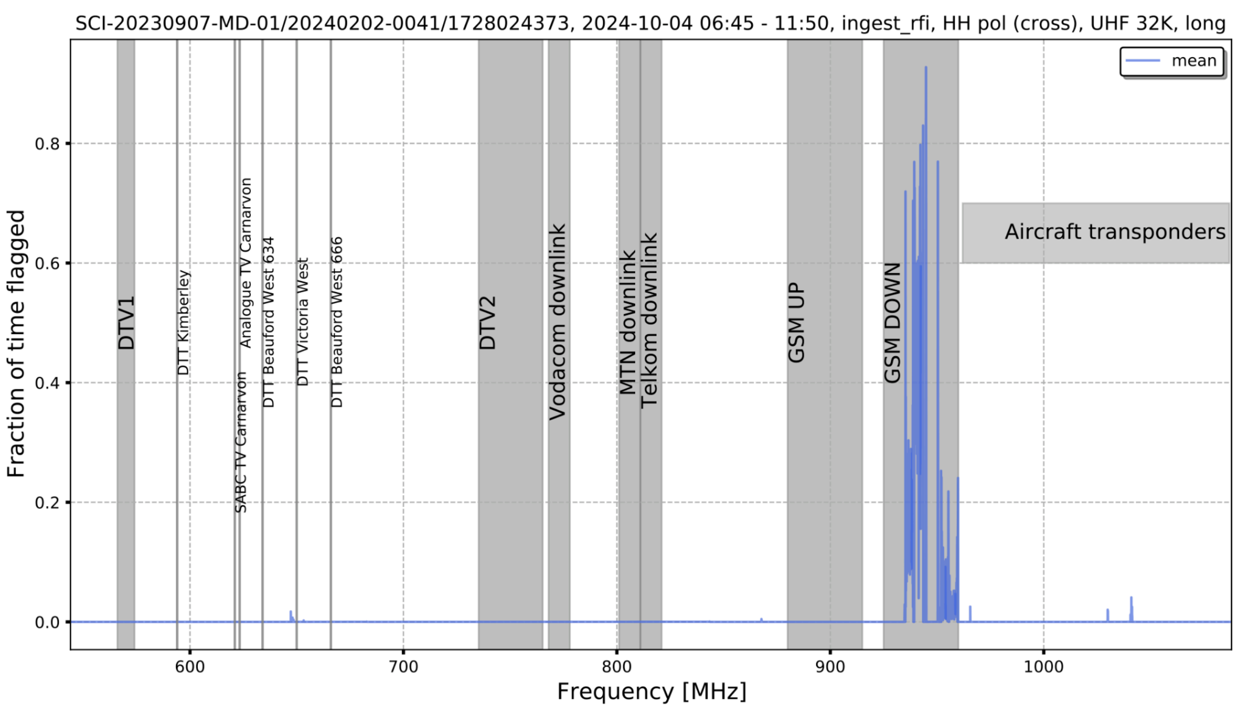

Figure 3: Mean of the cross correlation flags of long baselines of SB_ID:20240202-0041 dataset.

b. 2024, 08 October

The dataset is from an observation carried out on the 8th of October 2024. It is apparent that the correlation products exhibit patterns that are similar to those of the dataset in the previous section. The auto and short baseline cross correlations are contaminated by relatively weak RFI, but with a number of noisy channels smaller than that of SB_ID:20240202-0041 dataset.

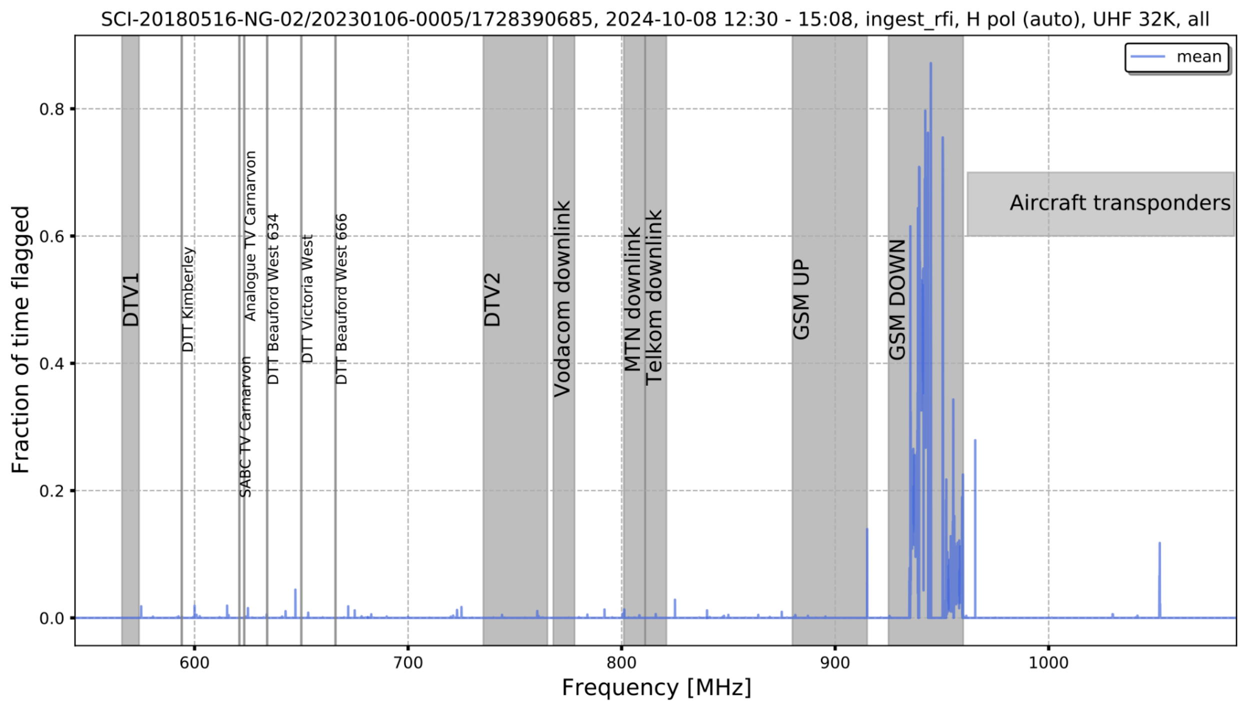

Figure 4: Mean of the autocorrelation flags of SB_ID:20230106-0005 dataset.

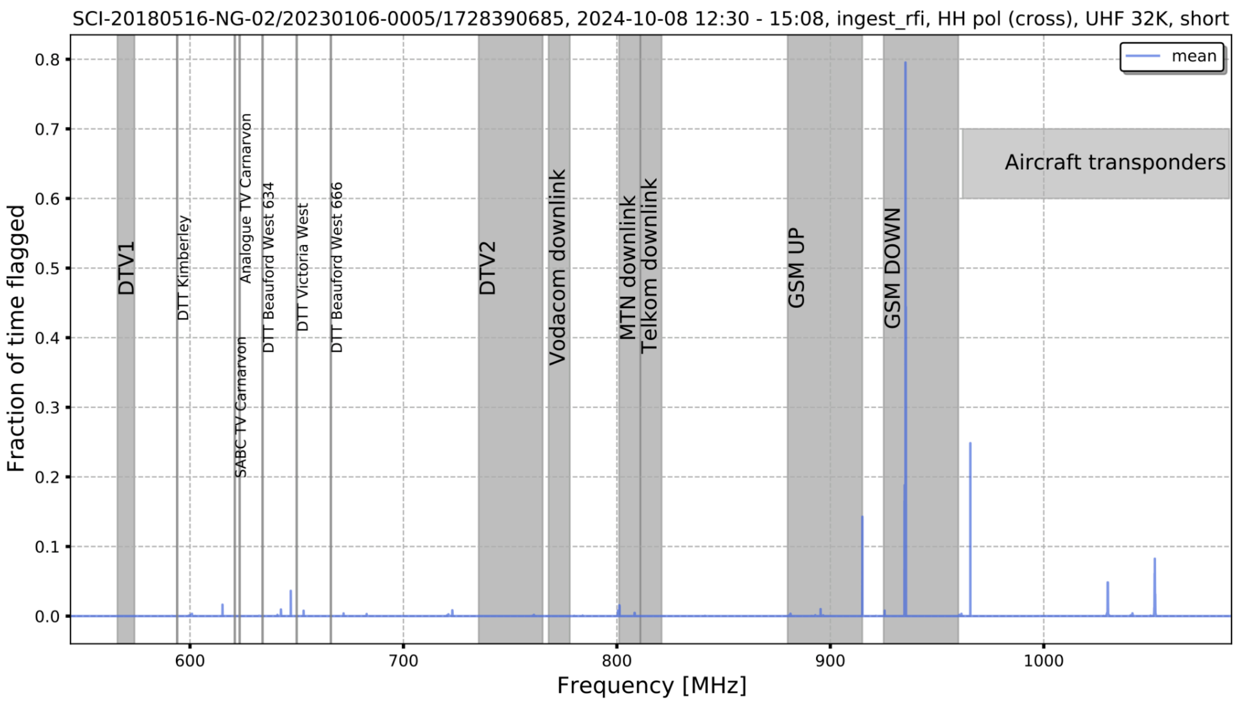

Figure 5: Mean of the cross correlation flags of short baselines of SB_ID:20230106-0005 dataset.

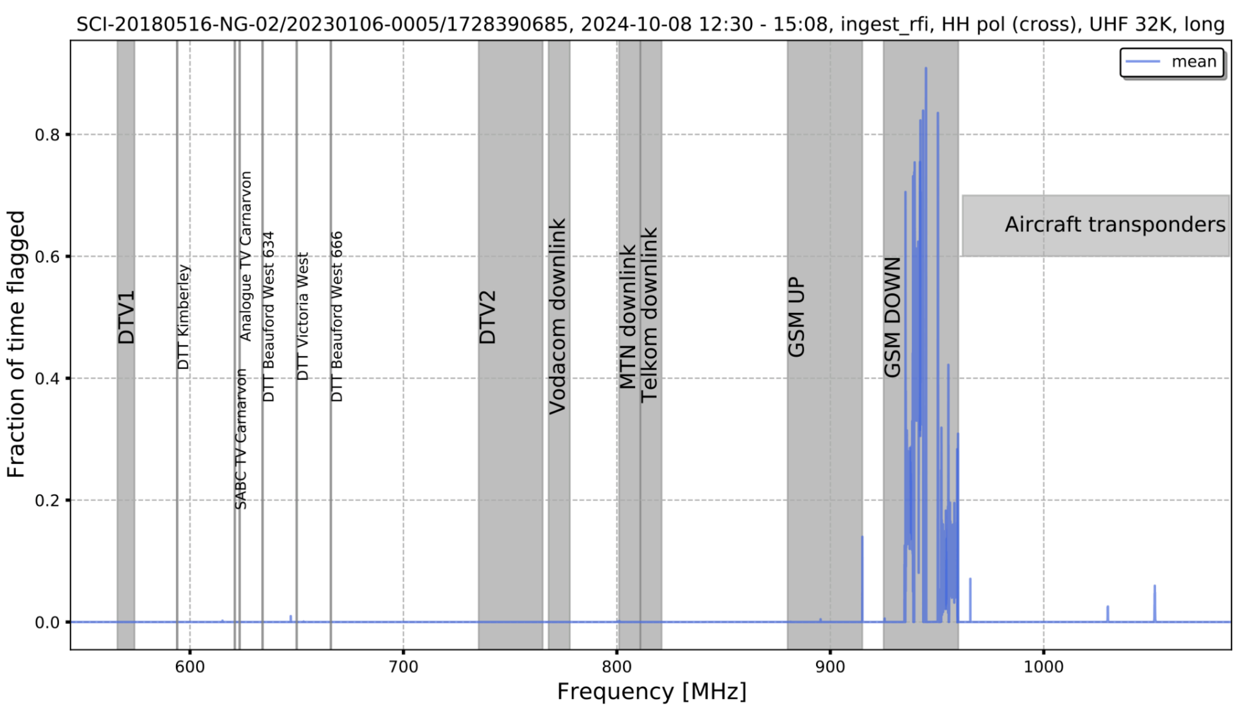

Figure 6: Mean of the cross correlation flags of long baselines of SB_ID:20230106-0005 dataset.

c. 2024, 18 October

The flags plotted in Figures 7, 8 and 9 are obtained from an observation carried out on the 18th of October. It can be clearly seen that in all three correlation products there is a relatively small sub-band at 811 - 821 MHz which is contaminated by RFI, indicating that all antennas are affected. This RFI emission, which is related to Telkom downlink according to recent investigation of the RFI team, is not supposed to be seen on site. Interestingly, relatively weak interference, with comb-like patterns, is noticed across the entire band in both auto and short baseline cross correlations (Figures 7, 8), unlike the long baseline cross correlation which appears cleaner (Figure 9). This clearly suggests that only short baseline antennas are contaminated by that noise.

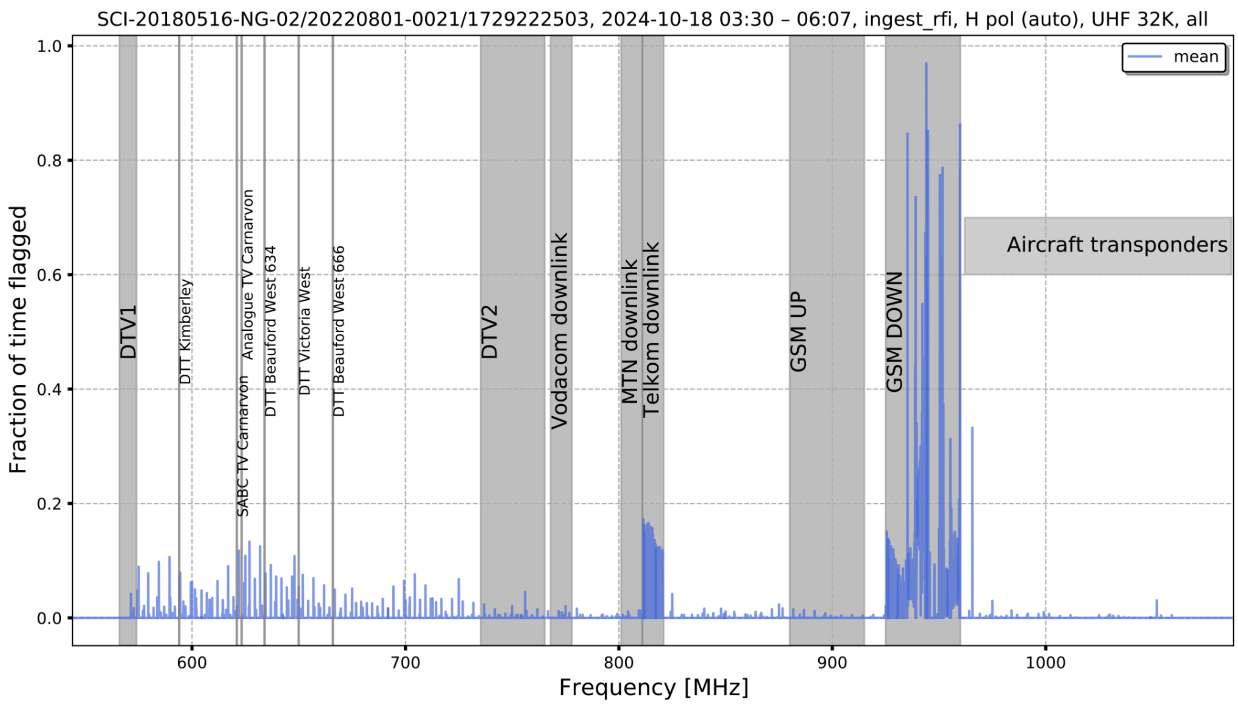

Figure 7: Mean of the autocorrelation flags of long baselines of SB_ID:20220801-0021 dataset.

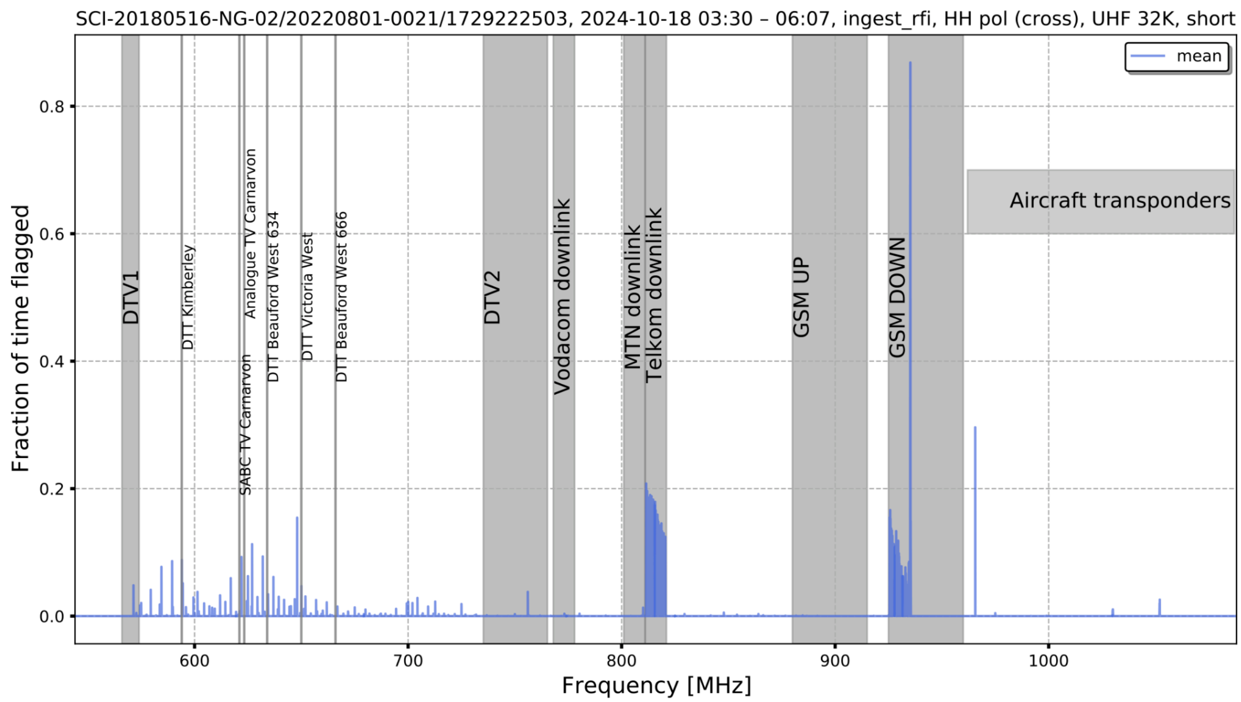

Figure 8: Mean of the cross correlation flags of short baselines of SB_ID:20220801-0021 dataset.

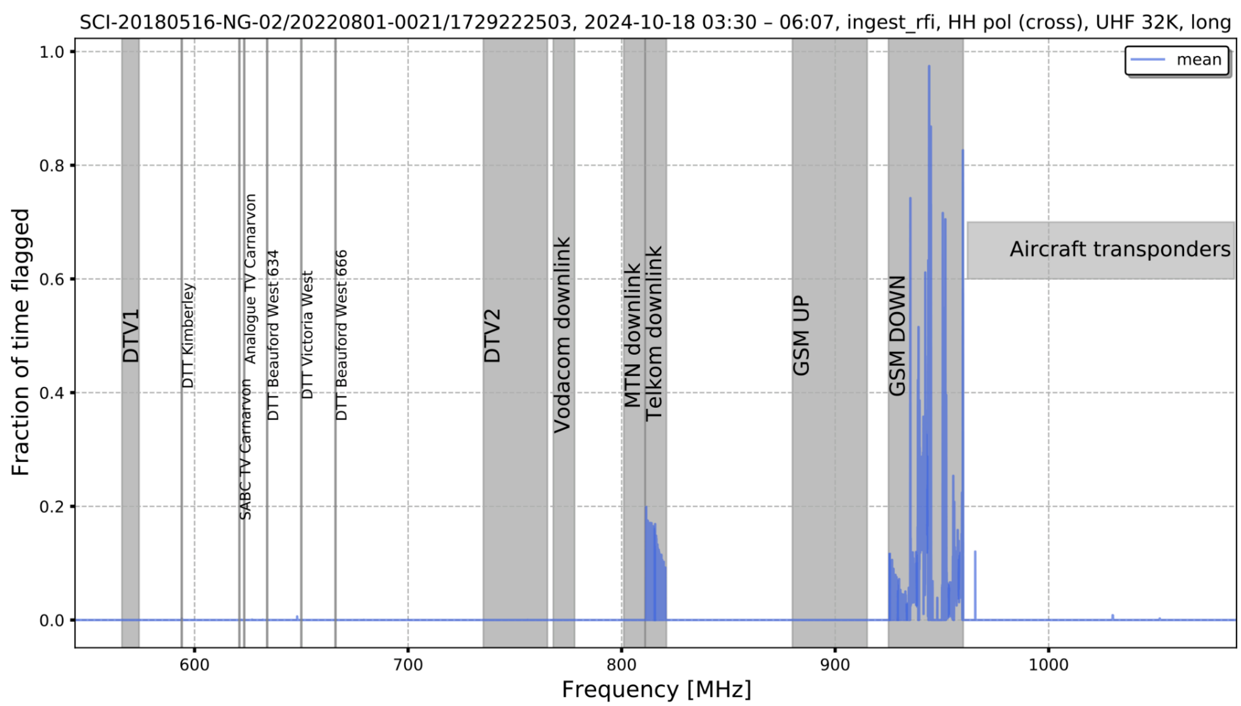

Figure 9: Mean of the cross correlation flags of long baselines of SB_ID:20220801-0021 dataset.

d. 2024, 22 October

The contaminants with comb-like patterns, which mainly affect the lower part of the band, are clearly noticeable in both the auto and short baseline cross correlations, although less pronounced in the latter, as shown in Figures 10 and 11. However the long baseline cross correlation appears to be clean, which points to the fact that the antennas at/around the core are more prone to the comb-like noise.

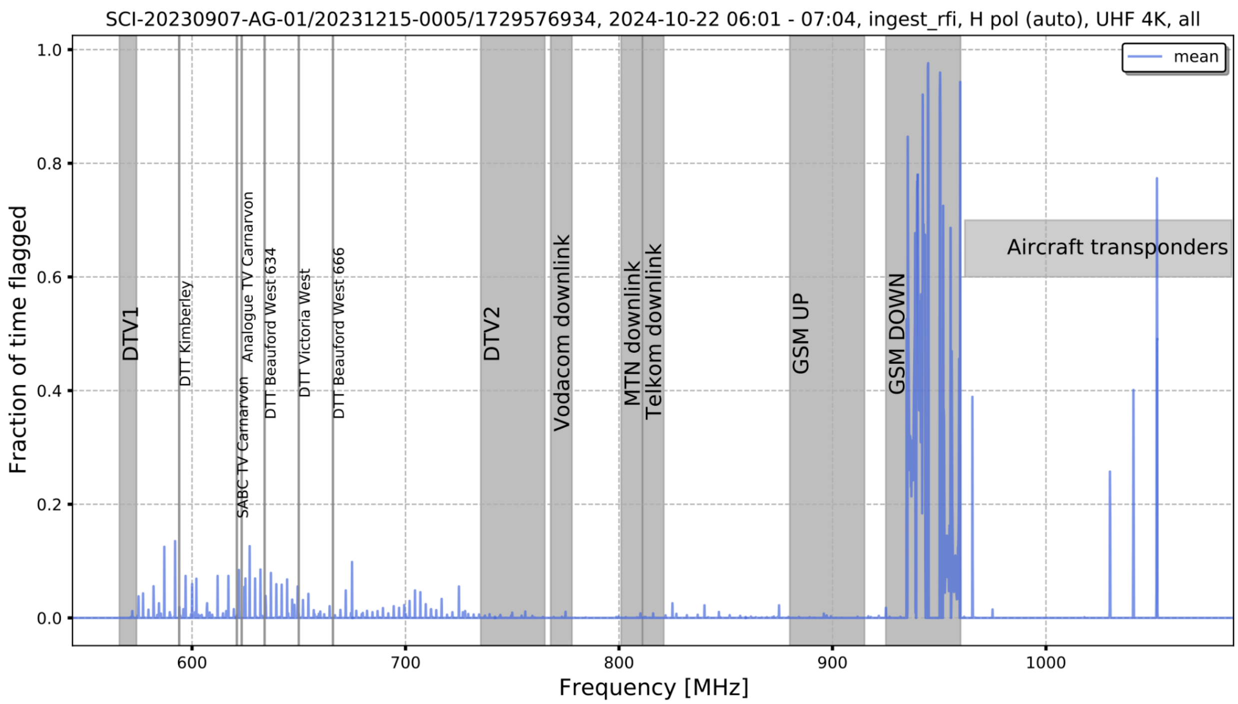

Figure 10: Mean of the autocorrelation flags of SB_ID:20231215-0005 dataset.

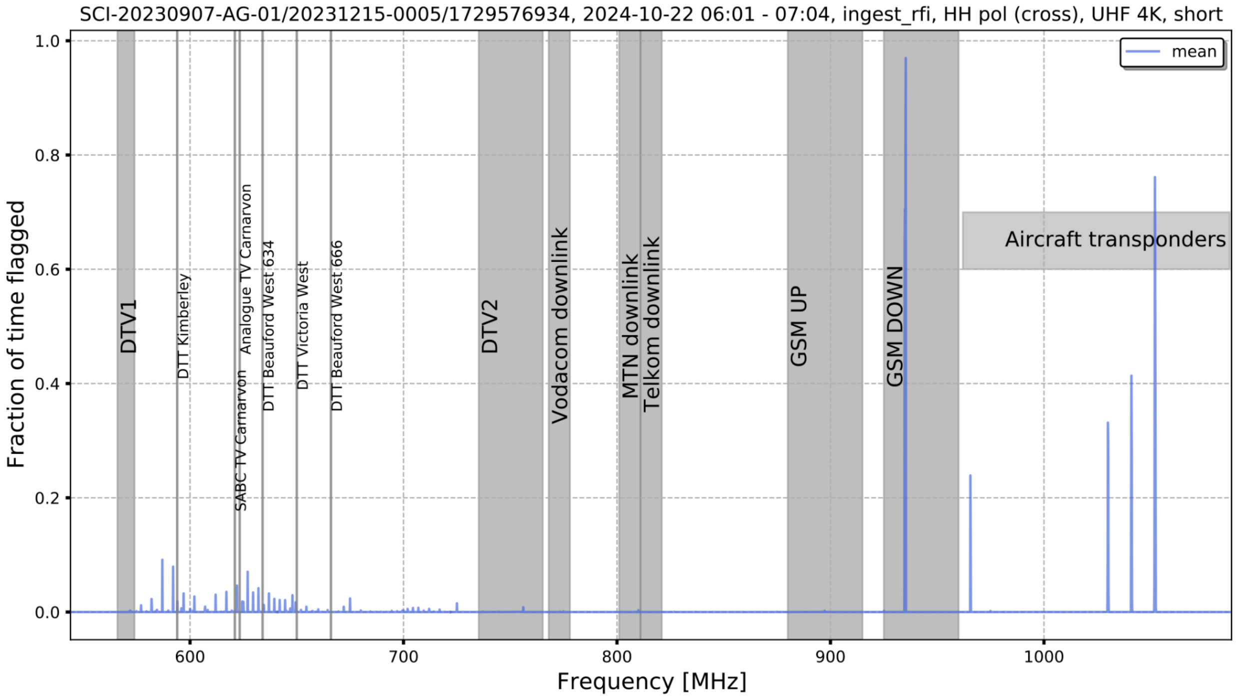

Figure 11: Mean of the cross correlation flags of short baselines of SB_ID:20231215-0005 dataset.

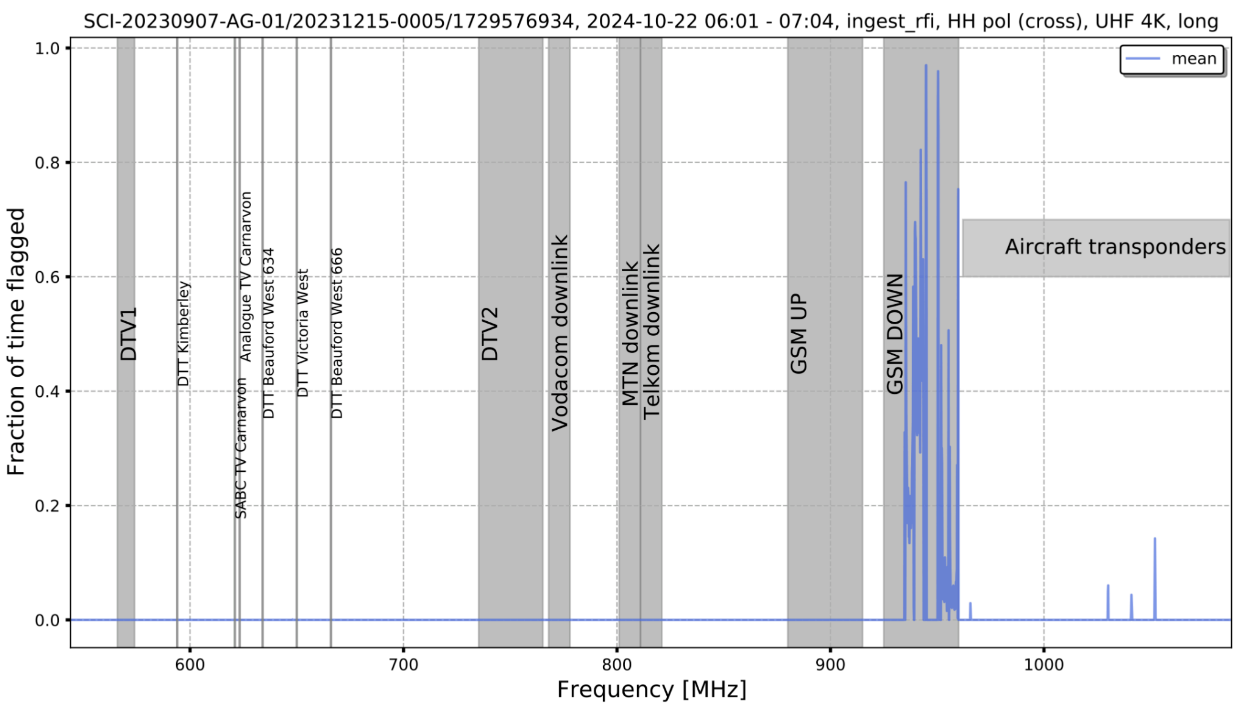

Figure 12: Mean of the cross correlation flags of long baselines of SB_ID:20231215-0005 dataset.

e. 2024, 27 October

The correlation products, shown in Figures 13, 14 and 15, are reasonably clean although weak contaminants are still noticeable in the autocorrelation. These weak RFI signals will not affect the data products for science as they are not visible in the cross correlations.

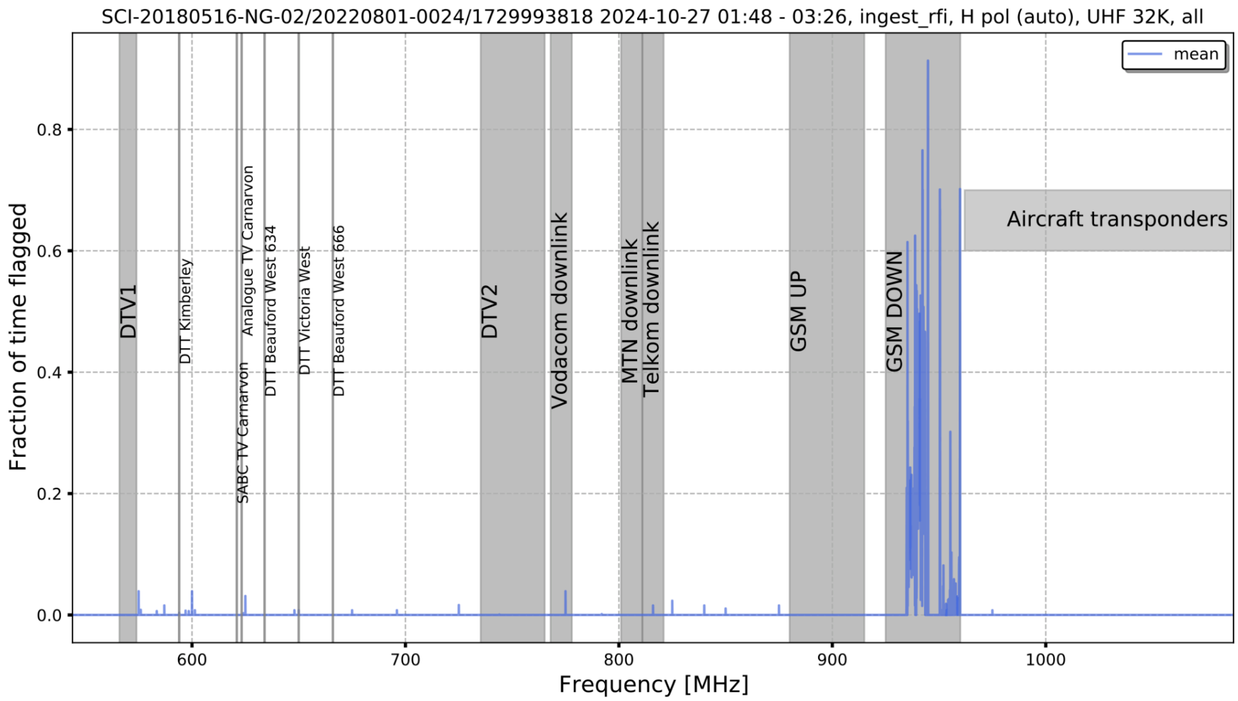

Figure 13: Mean of the autocorrelation flags of SB_ID:20220801-0024 dataset.

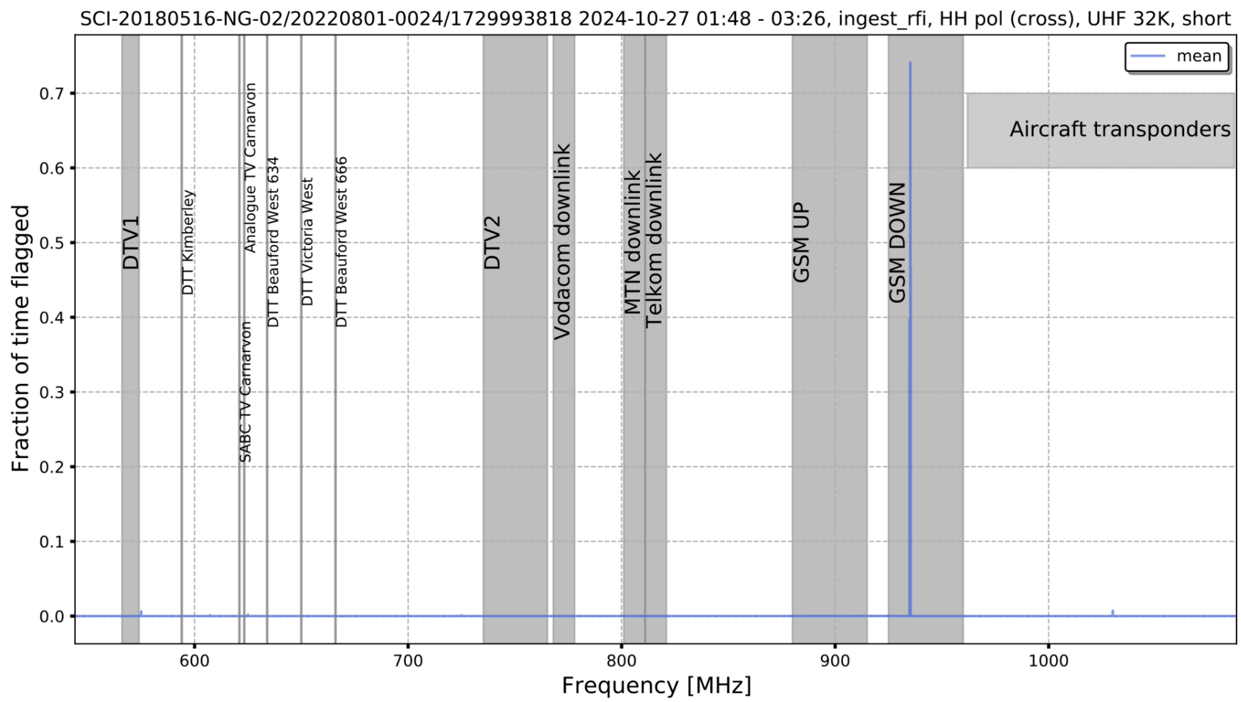

Figure 14: Mean of the cross correlation flags of short baselines of SB_ID:20220801-0024 dataset.

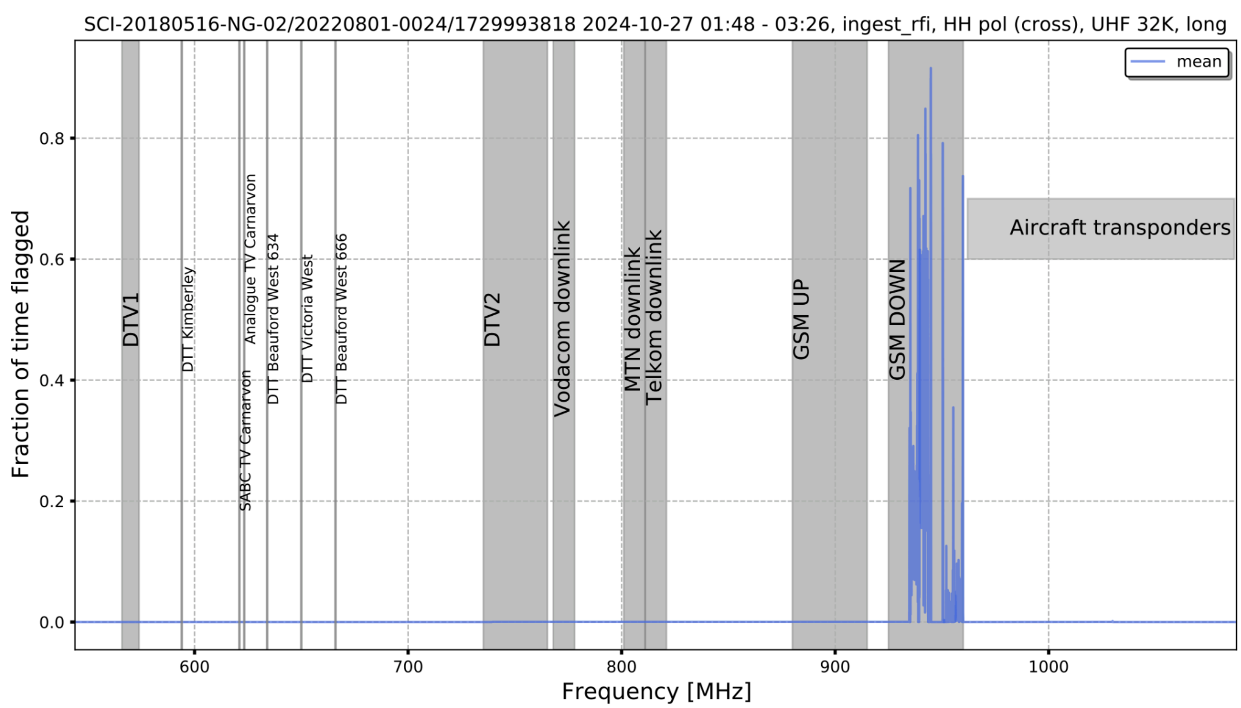

Figure 15: Mean of the cross correlation flags of long baselines of SB_ID:20220801-0024 dataset.

4 . RFI summary for October 2024 at UHF, HH cross polarization

Now we look at the overall RFI for the month of October by concatenating all science observations for the month. Here we select particularly 8s observations, observed at 4k or 32k channel mode. The monitoring and analysis of RFI are presented using the KATHPRFI pipeline [1], highlighting the impact of construction on RFI levels.

KATHPRFI algorithm

The kathprfi essentially processes two numpy arrays, Master and Counter, to update the number of RFI points and observations per voxel which refers to 5-D array of Time, Frequency, Baseline, Elevation and Azimuth. It iterates over chunks of frequency indices, baseline indices, and time indices. For each combination, the algorithm updates the Master array by the corresponding ‘Good flags‘ value and the Counter array by 1. Good flags for a given observation are based on a flag type, e.g. if flag type is calibration RFI then target tags for that observation will be obtained from a complete calibration report. The output of this is an xarray dataset that contains arrays of these two arrays and the dimensions of interest as mentioned above.

Computation of marginal probability

The RFI probability is computed by using two arrays from the output Zarr file obtained from the kathprfi pipeline : ’Master‘ and ‘Counter‘. These arrays represent the number of RFI points and the total number of observations, respectively. The function sums these arrays along a specified dimension to get total counts and then calculates the probability as the ratio of these sums.

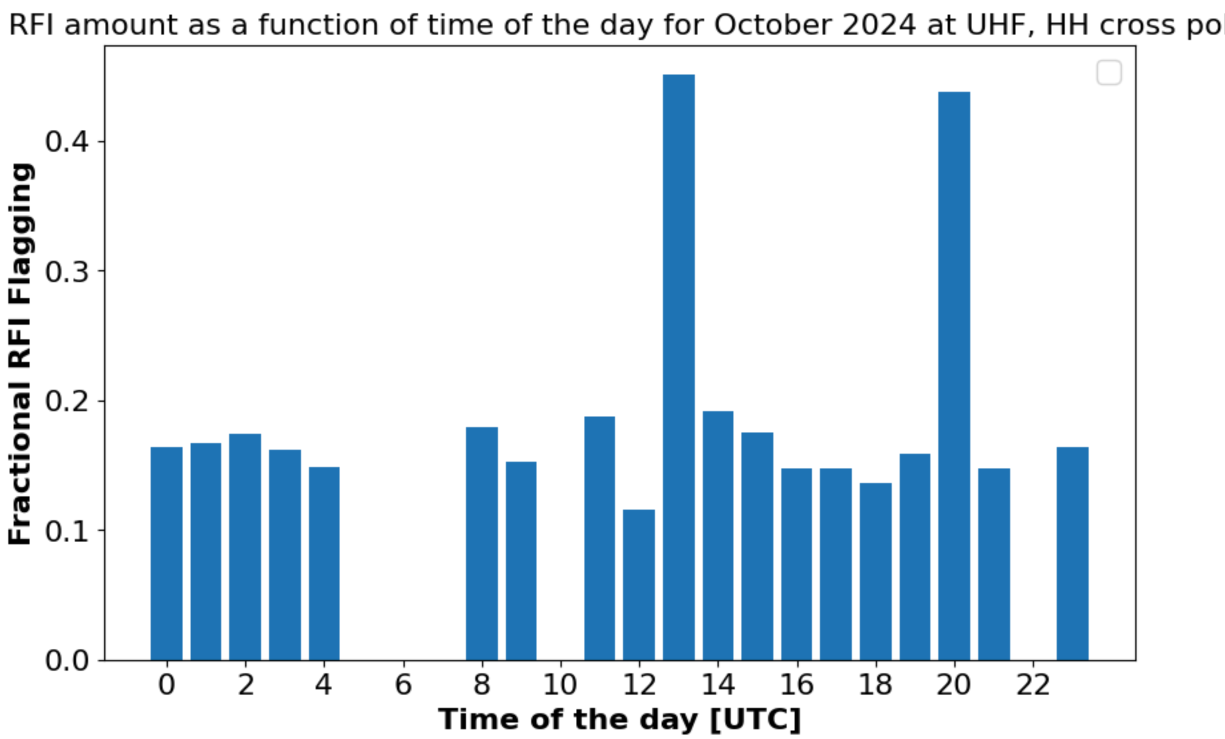

Figure 16: The plot above shows the fractional RFI flagging as a function of time of the day in UTC for October 2024 UHF horizontal cross pol observations.

Figure 16 illustrates the hourly behavior of RFI for the month of October 2024. This view allows us to identify specific time periods affected by RFI. The probability of RFI is relatively low most of the time, until the hours of 13:00 UTC and 20:00 UTC, the fractional RFI rises sharply to a maximum of 47% and 43% respectively. This suggests a period when RFI is especially prevalent. This could be attributed to a number of causes: This could be due to specific construction activities on site , also could be due to a source under observation at the time. To be able to characterize, we need to look at the frequency spectra at that time of the time, Figure 18 below answers this.

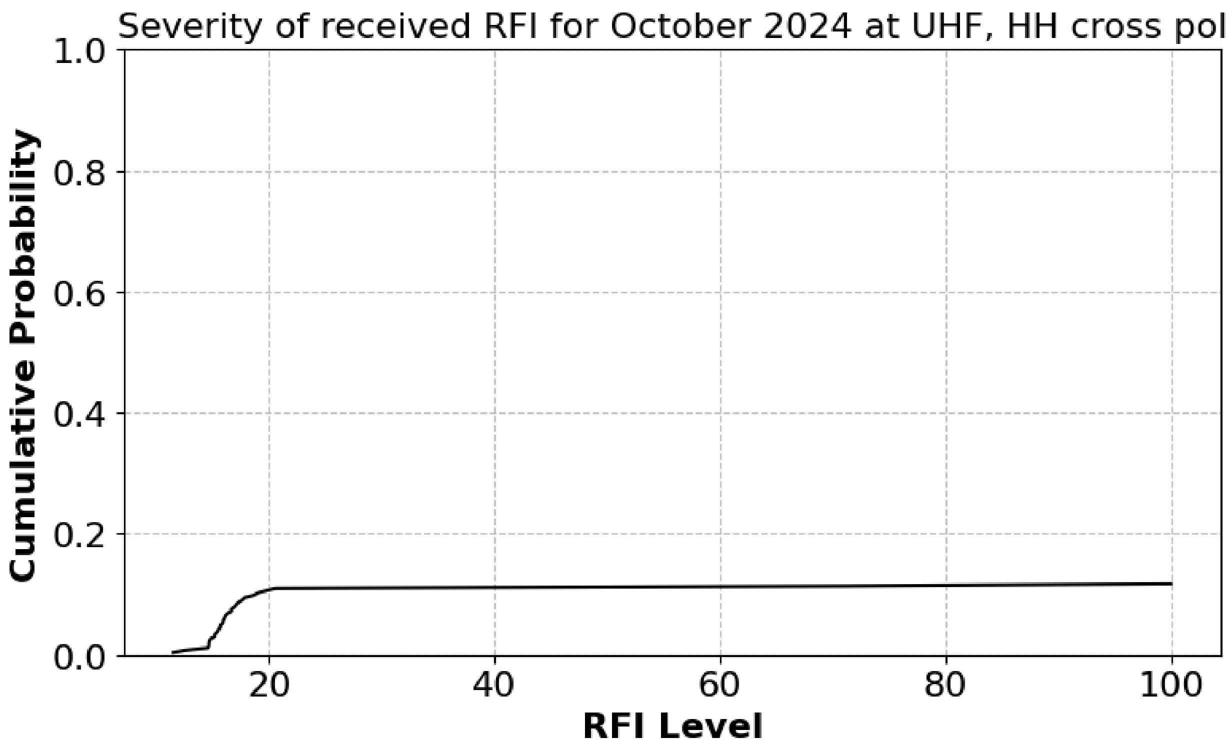

Figure 17: RFI distribution : cumulative probability of RFI as a function of RFI levels.

Figure 17, shows the Empirical Cumulative Distribution Function (ECDF) of RFI levels for October 2024. The x-axis represents the RFI levels, while the y-axis shows the cumulative probability, which represents the proportion of RFI data points that are less than or equal to each RFI level. This is useful for understanding the overall intensity distribution of the RFI events. For example, Figure 17 shows that at RFI level 20, around 12% of the data falls below this level with a steep slope from lowest probability, indicating that low RFI levels are more common or very concentrated in that range. Therefore, we can conclude that around 12% of all RFI data points fall at or below an RFI level of 20. The steep initial rise in the ECDF indicates that a large portion of RFI events occur at these lower intensities, while higher RFI levels are much less frequent, occurring only in a small fraction of the observations. Thus, severe RFI events are rare, but they do exist in the dataset.

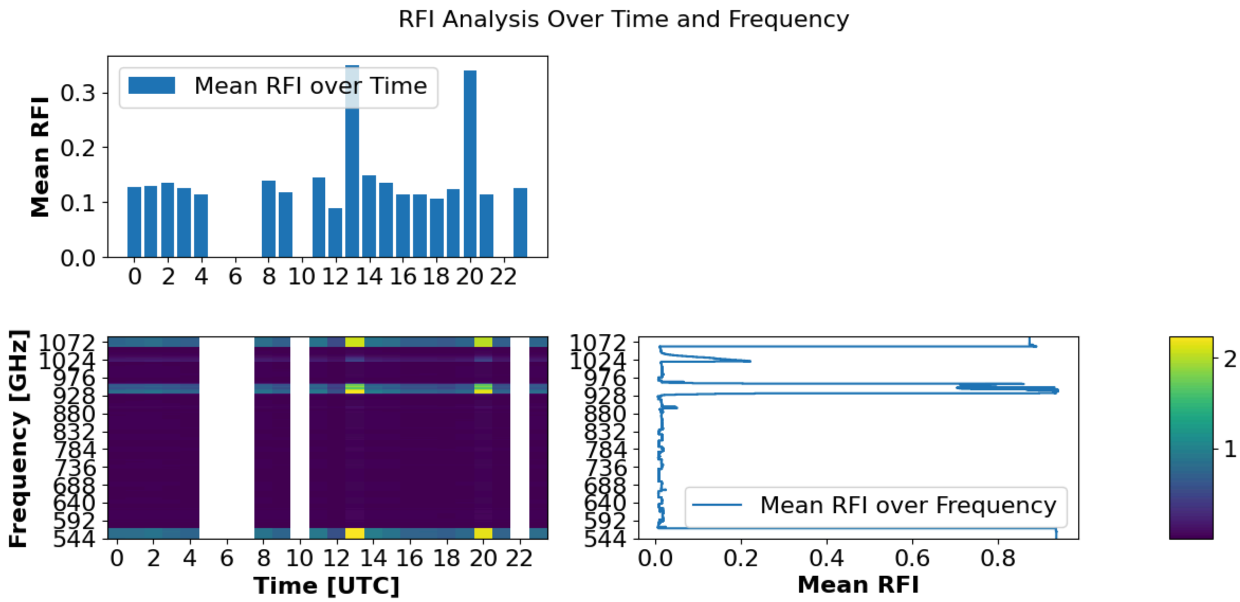

Figure 18: Mean RFI across time and frequency

Figure 18, depicts a 2 dimensional view of RFI patterns over time and frequency. The top 1D bar plot shows the median RFI levels over time exactly the same as Figure 16. The 2D heatmap in the centre represents the detailed RFI activity across time and frequency, where white areas suggest missing data or non-monitored frequency intervals. Notably, intense RFI zones appear as bright yellow or green patches as indicated by the colour bar (far right), notably between 544-592, 920–960 MHz and at higher frequencies above 1000 MHz. These strong yellow patches are seen at 13:00 UTC and 20 UTC and throughout other hours of the day RFI levels are moderate. The 1D plot on the right highlights mean RFI levels over different frequencies, with spikes indicating strong interference at specific frequencies. Fortunately we know the RFI present in these frequencies, GSM downlink – mobile communications.